Patrick Kelly

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EDUCATION

Ph.D.	Chemical Physics, University of Maryland, College Park	2022 - 2028 (expected)
M.S.	Physics, California State University, Fresno	2022
B.A.	Liberal Arts, St. John's College	2016

AFFILIATIONS/WORK EXPERIENCE

Research Assistant Roy Biology Group,
University of Maryland, College Park (Fall 2025 - Present)

Research Assistant Papoian Computational Biophysics Group,
University of Maryland, College Park (Spring 2023 - Present)

Research Rotation Jarzynski Group,
University of Maryland, College Park (Spring 2023)

Research Rotation Tiwary Group,
University of Maryland, College Park (Summer 2022 - Fall 2022)

Research Assistant Vitali Computational Condensed Matter Group,
California State University, Fresno (Fall 2018 - Spring 2022)

Research Assistant Strongly Correlated Electron Lab,
California State University, Fresno (Summer 2017 - Summer 2018)

Teaching Assistant Department of Chemistry,
University of Maryland, College Park (Fall 2024)

Teaching Assistant Department of Physics,
University of Maryland, College Park (Spring 2023, Spring 2024)

Teaching Assistant Department of Physics,
California State University, Fresno (Fall 2018 - Spring 2020)

Teaching Assistant Senior Laboratory,
St. John's College, Annapolis (Fall 2015 - Spring 2016)

Hodson Trust Summer Intern Delwiche Molecular Systematics Lab,
University of Maryland, College Park (Summer 2015)

RESEARCH AREAS

Computational biophysics; chemical physics; statistical mechanics; stochastic processes; condensed matter physics; quantum many-body physics; quantum fluids; classical simple liquids

GRANTS AND AWARDS

- Chickasaw Nation Tribal Scholarship Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, Fall 2025, Fall 2026
- Chickasaw Nation Higher Education Grant Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, Fall 2025, Fall 2026
- CMNS Diversity fellowship Spring 2022
- Chickasaw Nation Degree Completion Incentive Spring 2022
- Graduate Student Research and Creative Activities Award Spring 2022
- Faculty Sponsored Student Research Award Fall 2021
- Clay Science Education Scholarship 2021-22
- Louise and Dick Avakian Scholarship 2021-22
- Jean S. Hirayama Scholarship 2021-22
- James T. Shockley Scholarship 2021-22
- Jon R. Dews Physics Research Scholarship 2021-22
- Central California Research Symposium (CCRS) Third Place Graduate Oral Presentation Award 2020

SERVICE AND SOCIETY MEMBERSHIP

- University of Maryland, College of Mathematical and Natural Sciences Graduate Student Council, chemical physics program representative Fall 2023 - Present
- Physics Outreach Volunteer Fall 2017, Spring 2018, Spring 2020
- Information Theory Study Group Fall 2019

PUBLICATIONS

1. Zachary Gao Sun, Nathan Zimmerberg, Patrick Kelly, Carlos Floyd, Garegin Papoian, and Michael Murrell, “Feedback between F-actin organization and active stress governs criticality and energy localization in the cell cytoskeleton”, (*Nat. Phys.* ***21****, 1290–1302 (2025)*).
2. Christian Apostoli, Patrick Kelly, Annette Lopez, Kaelyn Dauer, Gianluca Bertaina, Davide Emilio Galli, and Ettore Vitali, “Spectrum of density, spin and pairing fluctuations of an attractive two-dimensional Fermi gas”, (*Phys. Rev. A* ***110****, 033306 (2024)*).
3. Annette Lopez, Patrick Kelly, Kaelyn Dauer, and Ettore Vitali, “Fermionic Superfluidity: From Cold Atoms to Neutron Stars”, (*Eur. J. Phys.* ***43****, 6 (2022)*).
4. Patrick Kelly and Ettore Vitali, “On the Accuracy of Random Phase Approximation for Dynamical Structure Factors in Cold Atomic Gases”, (*Atoms* ***9****, 88 (2021)*).
5. Ettore Vitali, Patrick Kelly, Annette Lopez, Gianluca Bertaina, and Davide Emilio Galli, “Dynamical structure factor of a fermionic supersolid on an optical lattice”, (*Phys. Rev. A* ***102****, 053324 (2020)*).

TALKS/PRESENTATIONS

- “Modeling Stochastic Mesoscale Fluid-Structure Interactions” International Physics of Living Systems Conference, Ludwig Maximilian University, Munich, Germany August 2026.
- "Heterogeneous Catalysis at the Origin of Life on Wet and Icy Worlds” Interdisciplinary Summer Student Talks, University of Maryland, College Park June 2026
- "The Origin of Life on Wet and Icy Worlds” Student Physical Chemistry Talks, University of Maryland, College Park May 2026
- "Heterogeneous Catalysis at the Origin of Life: Insights from Ab Initio Molecular Dynamics and Enhanced Sampling Techniques” Student Physical Chemistry Talks, University of Maryland, College Park August 2025
- "Stochastic Hydrodynamics in the Cell” Physics of Living Systems - Student Research Network (PoLS-SRN) Talk, University of Maryland, College Park May 2025
- “Biophysical Simulations with Immersed Boundaries” Annual Burgers Symposia, University of Maryland, College Park October 2024.
- “Combining Chemical Reactivity with Mesoscale Hydrodynamics” American Conference on Theoretical Chemistry, University of North Carolina, Chapel Hill June 2024.
- “Methods for Combining Chemical Reactivity and Mesoscale Hydrodynamics” International Physics of Living Systems Conference, International Centre for Theoretical Physics, Trieste, Italy June 2024.
- “The Origin of Life in Alkaline Hydrothermal Vents” Biophysics Student Talks, University of Maryland, College Park April 2024.
- “Simulating Avalanche-like Dynamics in Actomyosin Networks” Cell Bio 2023, Boston December 2023.
- “Self-Organized Criticality and the Cytoskeleton” Biophysics Student Talks, University of Maryland, College Park October 2023.
- “Calculating Chemical Potential with the Widom Insertion Method” Chemical Physics Rotation Course Talk, University of Maryland, College Park May 2023.
- “Long Short Term Memory for Molecular Systems” Chemical Physics Rotation Course Talk, University of Maryland, College Park November 2022.
- “Exploring the Quantum Many-Body Problem with Cold Atomic Gases” Physics Colloquium, California State University, Fresno April 2022.
- “Assessing the Accuracy of Random Phase Approximation for Dynamical Structure Factors in Cold Atoms with Quantum Monte Carlo” APS March Meeting March 2022.
- “Correlation Functions for Cold Atoms: Windows to the Invisible Quantum World” Physics Colloquium, California State University, Fresno December 2021.
- “Dynamical Structure Factors in Cold Atoms with Random Phase Approximation” American Physical Society (APS) Far West Sectional Meeting October 2021.
- “Random Phase Approximation for Cold Atomic Gases” Central California Research Symposium (CCRS) April 2021.
- “A Computational Study of Dynamical Properties of Cold Atomic Fermi Systems: Trapped Gases and Systems Moving on Optical Lattices” APS March Meeting March 2021.

- "Studying Cold Atomic Systems with Monte Carlo Simulation and Random Phase Approximation" APS Far West Sectional Meeting November 2020.
- "The Supersolid Phase in Simulated Cold Atomic Fermi Gases" CCRS April 2020.
- "The Dynamical Structure Factor of a Fermionic Super-solid on an Optical Lattice" APS Far West Sectional Meeting November 2019.
- "Exotic Phases in Cold Atoms: The Supersolid State" CCRS April 2019.
- "Method for the Growth and Stabilization of Rare Earth Nano-Particles" (co-author, poster) APS March Meeting March 2018.
- "Determining the Sequence of Cell Division in Spirogyra and Klebsormidium with Calcofluor White Staining" (poster) Phycological Society of America Annual Meeting August 2015.